Optics Communications 560 (2024) 130459

Contents lists available at ScienceDirect

Optics Communications

journal homepage: www.elsevier.com/locate/optcom

Optical encryption by dynamically modulating the spatial frequency of light fields

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ARTICLE INFO

Keywords:
Optical encryption
Dynamic modulation
Advanced coding technique

ABSTRACT

Optical encryption has received increasing attention recently as information security is playing a crucial role in the modern information society. Particularly, the intensive investigation and rapid development of metasurfaces have significantly facilitated the research of optical encryption with enhanced security and increased capacity for information. Nevertheless, most metasurface encryption schemes still suffer from the challenge of dynamic modulation. Here, we demonstrate a novel dynamic optical encryption scheme in which information can be encrypted in light fields in a reconfigurable manner. Specifically, the spatial frequencies of optical images are utilized as information carriers and dynamically modulated for encrypting information. With the developed reconfigurable Fourier optics technique, various kinds of messages can be directly embedded in the spatial frequency domain of the ciphertext images and maintained with high fidelity. Importantly, information security and capacity can benefit from high-speed dynamic modulation with advanced coding techniques, an example of which has been demonstrated via exploiting the position, profile, and time degrees of freedom of spatial frequency. The proposed novel optical encryption scheme may pave the way for realizing high-performance information encryption systems.

1. Introduction

Optical encryption is manifesting itself recently as one of the important techniques in information security since light has become a crucial physical platform for modern information systems [1]. Enhancing the security of information, such as private data and confidential documents, is important for both individuals and companies, which is also more and more challenging because the cracking ability is increased too. Various schemes of optical encryption have been explored by taking advantage of the unique electromagnetic property of light and its interaction with matter [2–6]. Particularly, metasurfaces have been employed in optical encryption to develop novel and powerful methods [7–9]. The reason is that metasurfaces can exotically manipulate light waves with numerous degrees of freedom by using artificial micro/-nanostructures to tailor the characteristics of electromagnetic waves, such as amplitude, phase, polarization, and angular momentum, among others [10–13].

Generally, the working principle of optical encryption powered by metasurfaces is that the secured information is encrypted on the metasurfaces and cannot be correctly retrieved unless specific light excitations are applied as the decryption keys. Considering practical realization schemes, metasurface color printing and metasurface hologram are the two typical and promising candidates among various techniques. For metasurface color printing, the secured information is usually images that are constructed by tailoring the scattering amplitude of light with metasurfaces, which is dependent on the wavelength, polarization, and other degrees of freedom of incident light [14–17]. For metasurface holograms, the secured information is also usually images, but the scattering phase of light is engineered by metasurfaces [18–20]. Although different properties of the scattered light are manipulated by metasurfaces, for both schemes, the main target is to multiplex information as much as possible in one single metasurface by exploiting different degrees of freedom as encryption channels [21–26]. In this way, the security of information is also enhanced. It should be noted that

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https://doi.org/10.1016/j.optcom.2024.130459
Received 2 January 2024; Received in revised form 5 February 2024; Accepted 9 March 2024
Available online 14 March 2024
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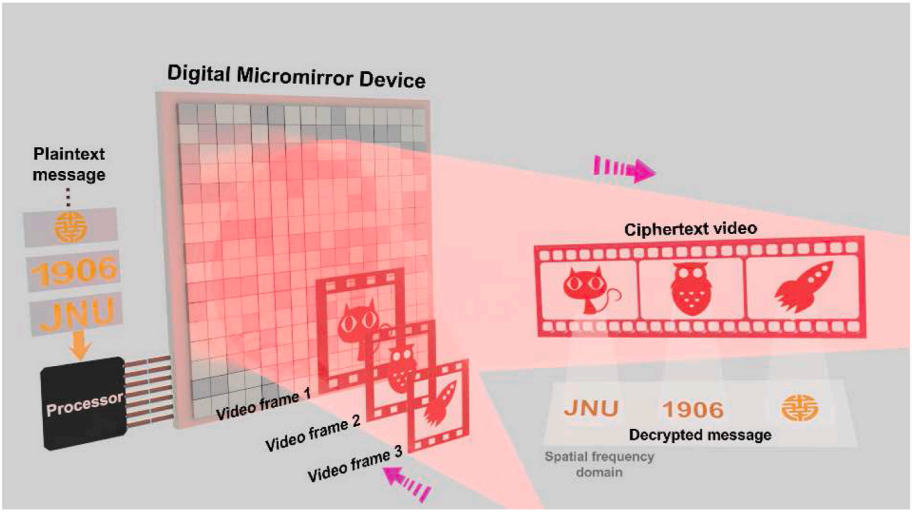


Fig. 1. Schematic of the dynamic optical encryption scheme. A laser beam with a sequence of video frames as the information carrier imping on the DMD that can dynamically modulate the SF of the light field, which is the principal process of the proposed dynamic optical encryption scheme. The plaintext messages after coding are fed to the processor of DMD to control the modulation. After being reflected by the DMD, the laser beam then contains the ciphertext video with the encrypted messages. The messages can be retrieved by examining the spatial frequency domain after applying the Fourier transform to the ciphertext video as the decryption process.

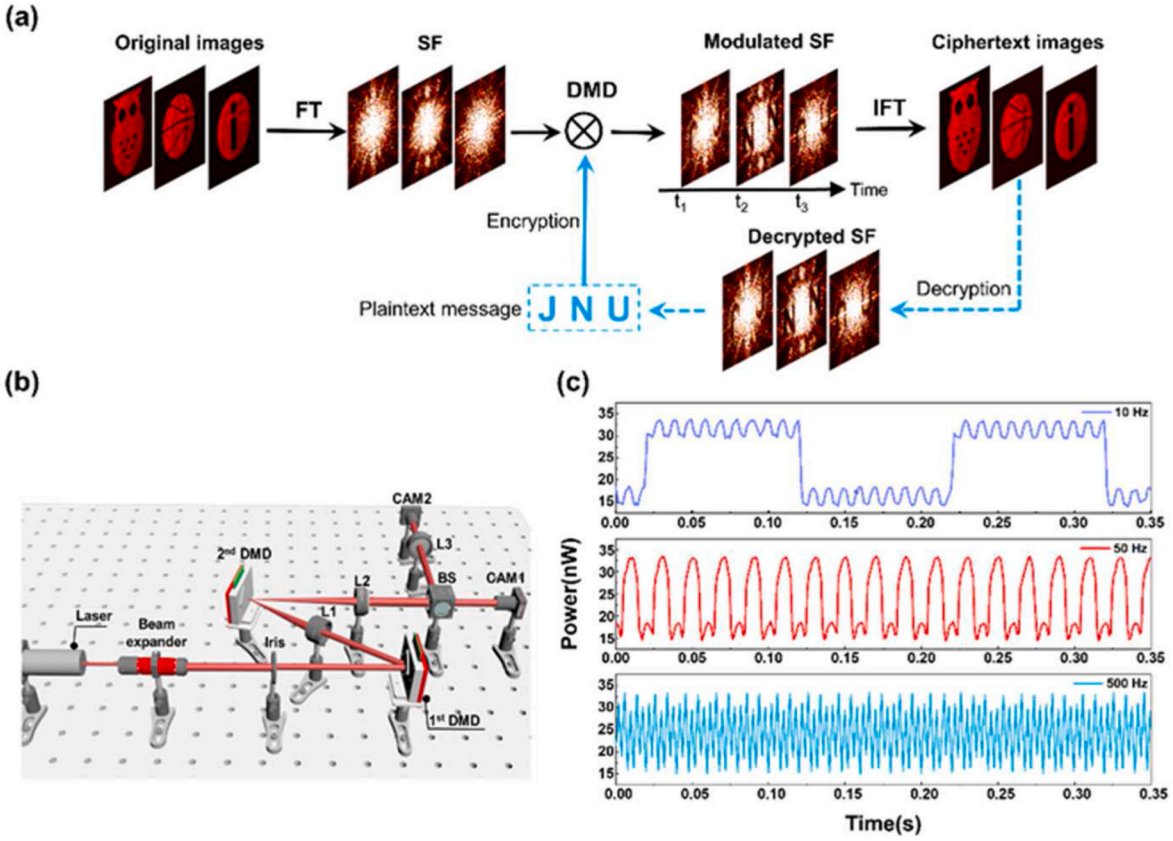


Fig. 2. Procedure and experimental setup of the dynamic optical encryption scheme. (a) Optical images as information carriers are first processed with the Fourier transform operation to enable the access of their SF. The plaintext message, for example, the letters of JNU, is dynamically loaded on the DMD to modulate the SF patterns for encryption. Finally, the modulated SF patterns are converted as ciphertext images with the inverse Fourier transform operation. Correspondingly, decrypted SF can be obtained from different ciphertext images through decryption. (b) Experimental setup of the RFO technique for implementing the proposed encryption scheme. L1 to L3, Lens; BS, beam splitter; Cam1 and Cam2, cameras. (c) The measured encryption rate of the proposed dynamic encryption scheme. The optical power of the modulated SF area as functions of time at different DMD updating rates of 10 Hz, 100 Hz, and 500 Hz. The low- and high-power levels indicate the on and off states of the DMD.

dynamic or reprogrammable metasurfaces provide a more promising platform that can further release the limited encryption channels of the static metasurfaces mentioned above [27–33]. Despite the brilliant progress in optical encryption by encrypting information on metasurfaces, some challenges are still to be addressed. For example, the hardware form of metasurfaces can greatly enhance information security because one must get the devices and set up particular decryption systems to obtain the information. However, this process also limits the efficiency of information transfer [34]. Another challenge is that the encrypted information is fixed and difficult to update once the metasurfaces have been fabricated, limiting the real-time applications once the decryption keys are leaked. The reprogrammable metasurfaces-based optical encryption can partially release the constraint mentioned above by updating the key with modulated incident light fields [29]. Nevertheless, it is still highly desired and challenging to realize an optical encryption scheme with strong security, high efficiency, and great flexibility, satisfying the growing demand for information security in future society. In fact, optical images can also serve as a promising platform for optical encryption, but the dynamic control is still to be addressed [35]. This method utilizes a hybrid optical-electronic encryption theme with high-speed modulation, which eliminates the issues of information processing efficiency and transfer.

Here, we propose and experimentally demonstrate a novel dynamic optical encryption scheme. Compared to the cases of encoding messages on physical devices, we prefer to encrypt messages on the spatial frequency (SF) domain of optical images by the reconfigurable Fourier optics (RFO) techniques based on digital micromirror devices (DMDs) modulation. Particularly, the process of message encoding can be dynamically controlled with the developed RFO techniques, in which the spatial frequency domain of optical images can be modulated by using reprogrammable DMDs. Therefore, on one hand, the information security can be increased in contrast to the case of directly concealing the optical images in the real domain. On the other hand, encrypted information can be flexibly changed at high speed, which enhances the transmission, recording, and decryption of information. We have experimentally demonstrated that encrypted information can be dynamically updated by programming the state profile of the DMD deployed in the RFO technique, while the output ciphertext images are nearly the same for different plaintexts. Importantly, the DMD can provide a promising opportunity for realizing flexible optical encryption by establishing advanced rules for encoding information. As an example, a solution of video encryption has been demonstrated by exploiting the position, profile, and time degrees of freedom for the SF of optical images with a dynamic modulation rate of 13 frames per second (fps). The dynamic encryption rate is also shown to be up to 500 Hz and is hoped to be faster.

2. Concept of the dynamic optical encryption

Fig. 1 schematically shows the concept of the proposed dynamic optical encryption scheme, in which light fields with dynamic images or videos as information carriers are directed to the DMD and reflected with the encrypted information as ciphertext video. To encrypt information, the plaintext messages are processed to control the states of micromirror units in the DMD, which then dynamically modulates the SF of the incident light field patterns. In this way, the information is embedded in the reflected optical videos and can be extracted by examining the corresponding SF. For example, a light field including a cat pattern within a beam is shined on the DMD as the information carrier. Then, a plaintext message of JNU, the abbreviated name of Jinan University, is fed to the processer of the DMD after proper coding. As a result, the DMD reflects parts of the incident light field at the Fourier domain, realizing the encryption of the "JNU" message in the SF of the cat image. Finally, the reflected light field of the modified cat image is outputted as the ciphertext video frame, which can be distributed and processed with great flexibility. Other plaintext messages can be similarly encrypted within different video frames. The receiver can obtain coded messages by directly identifying encrypted regions in the SF of ciphertext video frames on which the Fourier transform operation is digitally applied. Coded messages can be combined to obtain decoded messages according to the predefined code rules. More details can be found in section 5 about the encoded and decoded processes.

3. Working principle and experimental realization

To explain the concept of the proposed dynamic optical encryption scheme in detail, we schematically illustrate the basic procedure in Fig. 2(a) and describe the working principle in the following. First, the optical images are processed with the Fourier transform (FT), thus the corresponding SF can be accessed for encryption. Assuming that the electric field of the optical image is described as $E_{in}(x,y)$, the SF can be obtained with FT operation and expressed as $G(u,v) = \int\int_{-\infty}^{+\infty} E_{in}(x,y)e^{-i2\pi(ux+vy)}dxdy$. Then, the plaintext message is fed to the 2nd DMD directly or after coding to modulate the SF of the optical image. It is worth noting that information needs to be encrypted with central symmetry in the spatial frequency domain due to the application of Fourier transform operation. Therefore, the SF with encrypted information can be expressed as $G_m(u,v,t) = H(u,v,t)\cdot G(u,v,t)$, where $H(u,v,t)$ is the modulation function mainly accounting for the effects of DMD which enables the modulation of the SF of the input images. The modulation function $H(u,v,t)$ not only includes the planar position parameters u and v in the spatial frequency domain but also can be changed over time with the parameter of t to facilitate flexible encryption methods. The encryption speed is mainly limited by the modulation rate of the DMD, which is about 9 kHz in our case and could significantly increase the encryption efficiency compared to the metasurface schemes. Finally, the modified SF light fields are converted as ciphertext images by the inversed Fourier transform (IFT), and the output light field can be written as $E_{out}(x,y,t) = \int\int_{-\infty}^{+\infty} G_m(u,v,t)e^{i2\pi(ux+vy)}dudv$. Consequently, the plaintext message will not be displayed in the ciphertext images, realizing the information encryption (details of the simulation results of the process can be found in supplementary section 1). Further, the plaintext message can be retrieved by applying the FT operation on the ciphertext images and then analyzing the corresponding SF property.

We have developed the reconfigurable Fourier optics technique and built the corresponding experimental setup to realize the encryption procedure discussed above, which is shown in Fig. 2(b). A collimated laser beam (632.8 nm wavelength; DH-HN300, Daheng) is expanded to the entire active area of the first DMD (DLP6500, TI, resolution: 1920*1080, 8bit) in the object plane to generate the optical images as information carriers. The SF is obtained by applying the FT operation on the optical images with a lens (L1, f = 100 mm) and then projected on the second DMD (DLP6500, TI) for encryption in the Fourier plane. The modulated SF light fields reflected from the second DMD are converted as ciphertext images by a lens (L2, f = 100 mm) and finally recorded by a camera (CS165CU/M, Thorlabs, resolution: 1920*1080). Simultaneously, the modulated SF pattern can be monitored by another camera (CS165CU/M, Thorlabs) through a beam splitter (50:50, Daheng) and a lens (L3, f = 75 mm). In the proposed dynamic encryption scheme, the DMD is deployed for fast modulation of the SF, which also determines the dynamic encryption rate. To examine how fast can the proposed encryption scheme operate, we have directly measured the varying power of the modulated SF areas. As shown in Fig. 2(c), the time-varied power indicates that different dynamic encryption rates can be achieved by controlling the updating rate of the DMD. The low and high levels of power represent the on and off states of the DMD, respectively. Particularly, even at a rate of up to 500 Hz, the sharp contrast of the power levels can still be identified. It should be noted that the dynamical encryption rate is possible to be up to 9 KHz, namely the maximum modulation speed of the deployed DMD. Nevertheless, the response

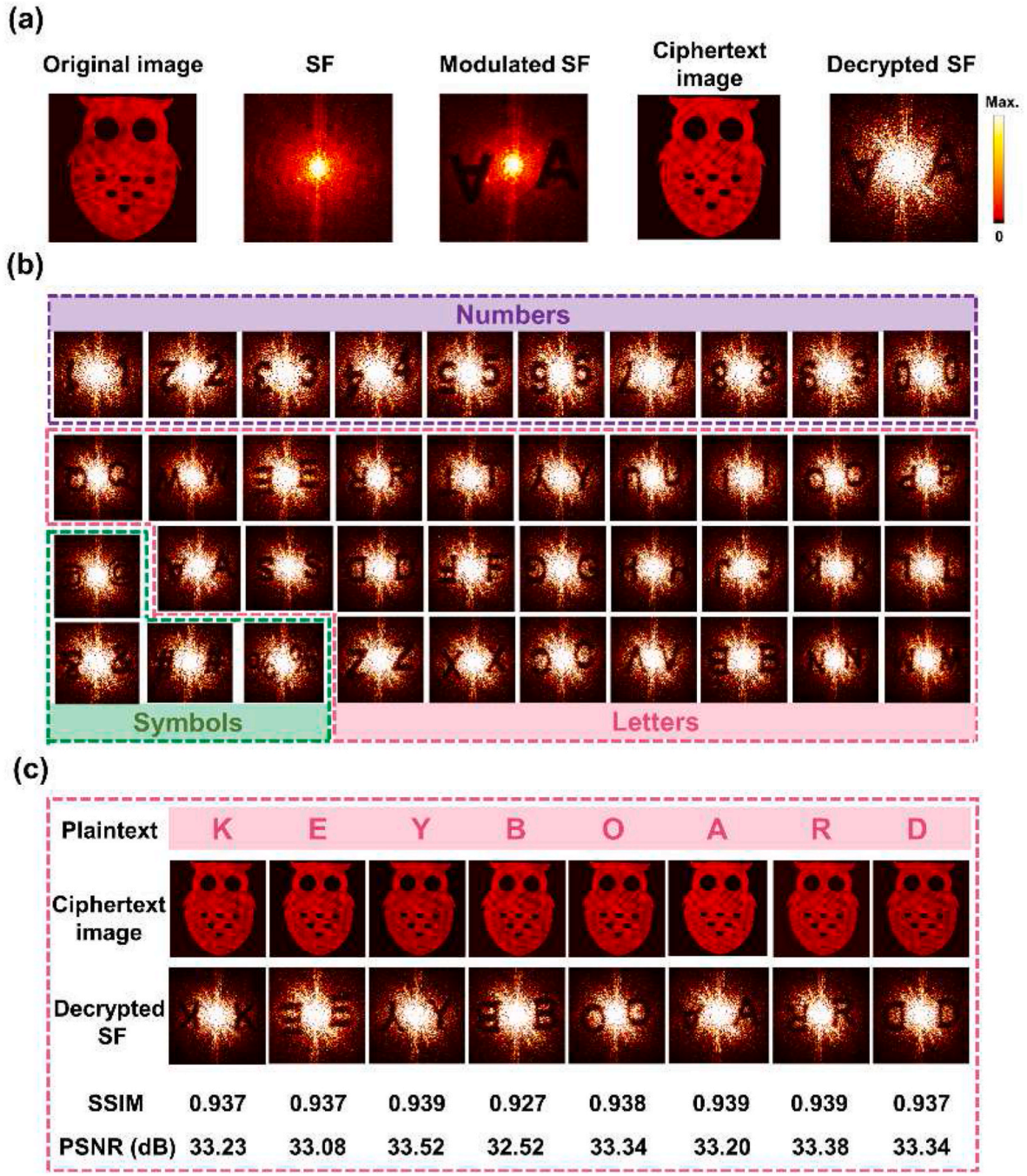


Fig. 3. Experimental results of the dynamic optical encryption. (a) From left to right: original optical image as the information carrier, intensity pattern of the SF, modulated intensity pattern after encryption, ciphertext image, and decrypted intensity pattern of the SF. (b) Decrypted intensity pattern of the SF from the ciphertext images of various letters, numbers, and symbols after encryption. (c) The word "KEYBOARD" as a plaintext message is encrypted with the same owl image in a sequence. The ciphertext images are nearly the same while the decrypted results show different letters in the intensity patterns of SF. All the SSIMs are about 0.93 and all the PSNR are about 32.2 dB, indicating that the ciphertext images can maintain the high quality of the original image.

speed of our optical power meter (S120C, Thorlabs) limits the measurable rate (details of the measured experimental setup can be found in supplementary section 2).

4. Message encryption in the spatial frequency of an optical image

With the experimental setup described above, we have verified the proposed dynamic optical encryption scheme, the results of which are presented in Fig. 3. For an efficient encryption method with high security, the secured information should be timely updated and encrypted with different representations. The DMD deployed in the experimental setup can exactly satisfy this requirement. We first demonstrate that a message can indeed be encrypted in the SF of an optical image. As shown in Fig. 3(a), an optical image of an owl pattern is considered as the information carrier. For such a pattern, the lower components of the SF behave as a light spot with some ripples around the central zero frequency in the intensity pattern. By controlling the DMD to disable the reflection of the light field for some parts of the SF, the letter A can be embedded in the intensity pattern. After converting

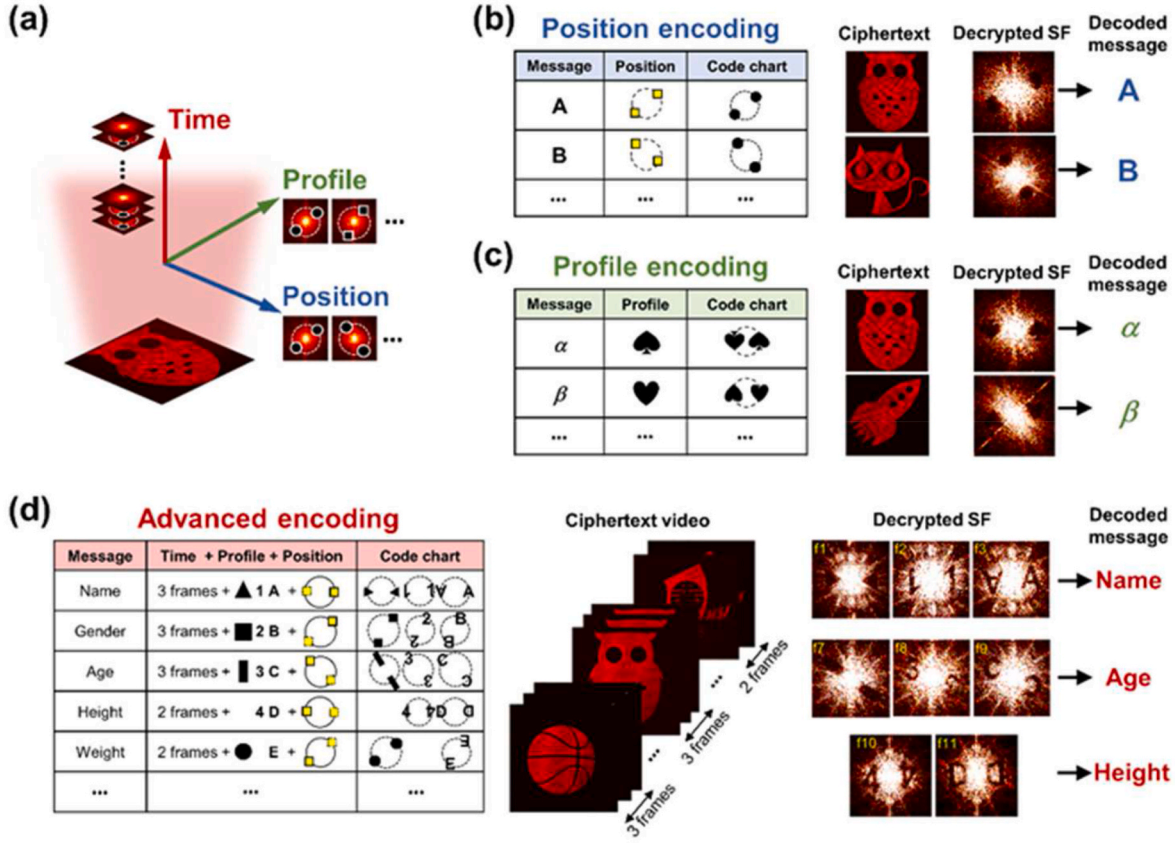


Fig. 4. Advanced coding techniques for the dynamic optical encryption scheme. (a) The position, profile, and time degrees of freedom of the SF of images are exploited for realizing advanced coding techniques. (b) Coding rules and experimental results for the encryption with the position dimension of the SF. (c) Coding rules and experimental results for the encryption with the profile dimension of the SF. (d) Advanced coding rules and experimental results for the encryption with the combination of the three dimensions of the SF.

the modulated SF back to the real image domain, a ciphertext image with the information of letter A can be generated. It can be found that the ciphertext image is nearly the same as the original image, and no feature of the letter A can be noticed, ensuring information security. The details of the image are slightly affected by the encrypted information but are hard to observe directly. Particularly, we have found that the structural similarity index measure (SSIM) characterizing the similarity between two images is about 0.93, indicating that even with partial modulation of SF, the ciphertext image can mostly maintain similarity with the original image. In addition, we also introduce the parameter Peak Signal-to-Noise Ratio (PSNR) to represent the quality of the ciphertext image, which is 32.23 dB (supplementary section 3). After decryption with the IFT applied to the ciphertext image, the encrypted information can be correctly identified, as shown in the last figure of Fig. 3(a).

Since the DMD can conveniently modulate the SF of optical images, the secured information can be easily updated on demand. As an example, we have further shown that different information can be separately encrypted in the same optical image dynamically. Fig. 3(b) summarizes the decryption results for 40 elements of a computer keyboard with the same owl image, which are retrieved from the experimentally obtained ciphertext images. It can be noticed that the corresponding shapes of numbers, letters, and symbols can be well recognized, indicating good quality for the encryption and decryption of the proposed method. Importantly, the ciphertext images for the 40 cases are nearly the same and no feature can be identified (supplementary section 4). For instance, we can encrypt the word "KEYBOARD" with the same image, and the results are shown in Fig. 3(c). Each letter of the word is dynamically encrypted in the SF of the same owl image in a sequence. The average SSIM index is about 0.93 and the average PSNR index is about 32.2 dB, which means the output ciphertext can be maintained with high similarity and high quality. In this sense, information security can be guaranteed. By applying the inverse Fourier transform which is also the key to the ciphertext images, analyzing the SF patterns, and combining the letters, the correct message can be retrieved. The ability to encrypt different information can find more useful applications in practice.

5. Advanced coding techniques for applications

Encrypting letters solely in the SF domain of optical images might be vulnerable for attackers, who can perform the FT operation on the ciphertext image, and then the messages would be obtained with careful analysis. Nevertheless, the proposed dynamic optical encryption scheme can leverage the flexibility and high speed of DMD to enhance security with advanced coding techniques for messages. Specifically, we have demonstrated that three degrees of freedom can be exploited for establishing advanced coding rules, namely, the dimensions of position, profile, and time, as shown in Fig. 4(a). For the encryption with the position dimension, it means that the different positions in the SF domain are modulated to represent the message according to specific coding rules. Here, the specific coding rules are the coding protocols that different combinations of geometric shapes and their positions at the spatial frequency domain as well as their changes over time can be employed to code particular messages. For instance, as shown in Fig. 4 (b), considering the SF in the polar coordinate, the radial and angular degrees of freedom can be employed for encryption. Particularly, the SF components at different angles are erased to represent the message of

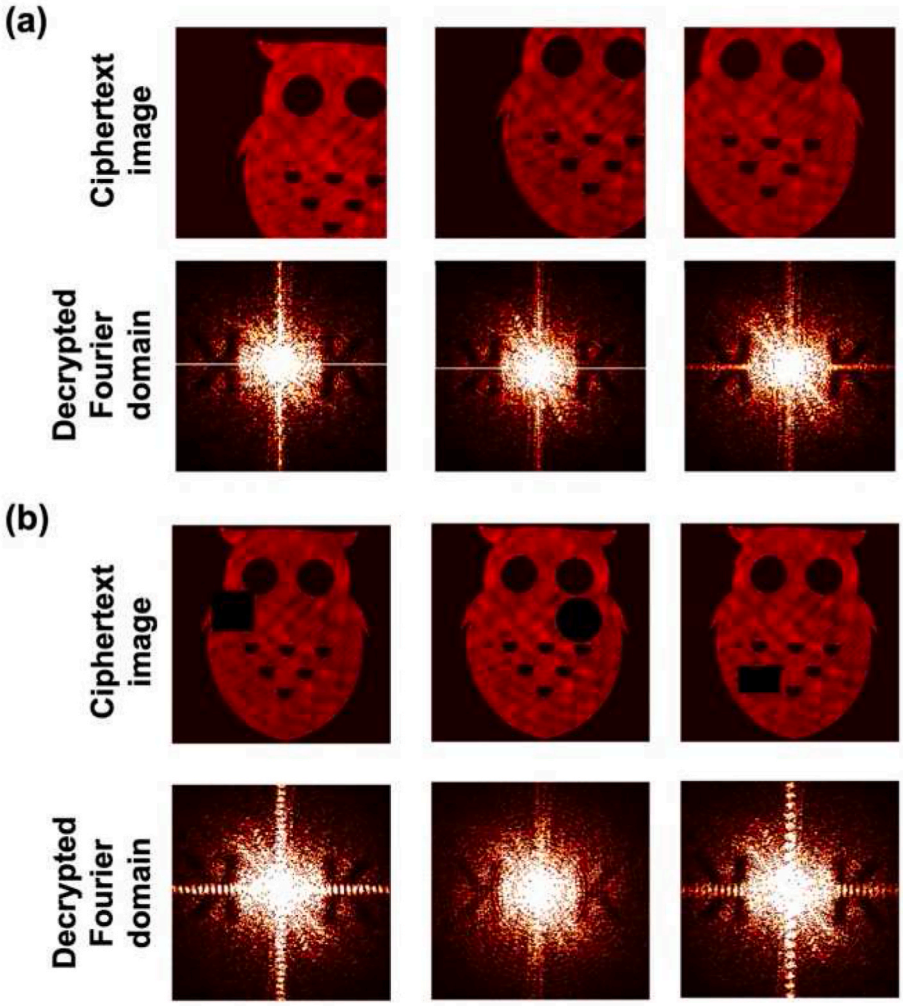


Fig. 5. Robustness of ciphertext images against the image quality degradation. (a) Partial region of the ciphertext images is intentionally cropped and the decrypted information can be recognized in the decrypted SF intensity patterns. (b) Different artifacts at different locations of ciphertext images and the decrypted information can also be recognized in the decrypted SF intensity patterns.

two letters A and B. Besides the position dimension, the areal profile of the SF components is another dimension for encryption. For the profile dimension shown in Fig. 4(c), although the position of the modulated SF is nearly the same, the areal profiles can be different to represent different messages. For example, the spade and heart patterns of the erased SF indicate the message of two Greek letters α and β.

To take it a step further, the dynamic controlling ability for SF modulation in the proposed encryption scheme adds the time dimension on the previous two degrees of dimension for realizing more powerful coding rules. For the time dimension, the message can be encrypted in an optical video consisting of several image frames. Therefore, the combination of the three dimensions can enable complex message encryption, as illustrated in Fig. 4(d). Taking personal information as an example, the name, gender, and other information can be encrypted with different combinations of position, profile, and time variables of the SF. For example, the name information can be coded with 3 frames of a video. In these frames, the triangle shape, number 1, and letter A are encrypted at a similar position of the SF to represent a particular name. Finally, all the frames constitute the ciphertext video (More details can be seen in **Visualization 1**). To retrieve the secured information, the authorized one who gets the code chart can decrypt the ciphertext video frame by frame and decode the frame sets (supplementary section 5). It should be noted that adding captcha codes (Eg. Adding a special string of numbers as captcha codes during the process of coding) is permitted to ensure the security of the encoded message.

6. Robustness analysis

In practical information transport channels, it is usually encountered that the image quality will be degraded, or even some artifacts will be introduced. In this case, we should consider whether the encryption can be robust against channel degradation. To simulate this situation, we intentionally modified the ciphertext images and examined the decryption results. As shown in Fig. 5(a), we considered the case that only parts of the ciphertext image are obtained, which could occur when the capturing camera is perturbated. It can be found that the letter is correctly retrieved after decryption. We have also considered another case in which some geometric artifacts are introduced to the ciphertext image. As shown in Fig. 5(b), the extraction of the correct letter is also possible. In this case, the geometric artifacts have sharp horizontal and vertical edges, resulting in cross-shaped bright lines in the decrypted Fourier domain images. Although these bright lines will affect the image quality, the decrypted results can still be obtained. Therefore, the results show that ciphertexts from the proposed optical encryption method are robust against particular attacks on ciphertext images. It is worth noting that ciphertext images are still robust under more serious attacks (supplementary section 6). This can be attributed to the fact that the SF is related to the whole image other than local features. Since the

information is accompanied by the whole image, the sensitivity of the information to the local changes of the environment is weak, which means that the encrypted information is robust to attacks.

7. Conclusion

In this work, a novel optical encryption scheme has been proposed and experimentally demonstrated to dynamically encrypt information in the SF domain of optical images and videos. The ciphertext generated with this scheme is contained in the images and videos, which can be recorded as digital files and thus are highly compatible with modern information processing technology, facilitating immediate applications in current systems. It should be noted that similar encryption methods can be developed digitally and realized with electronic computers and programs. Nevertheless, the proposed scheme can take great advantage of optical technologies, such as parallel and high-speed information processing ability, benefiting real-time and high throughput applications. Meanwhile, considering the metasurfaces encryption scheme, the metasurfaces are the ciphertext carriers and the fabrication quality is also critical since some small imperfections might induce incorrect results in the decryption process. In contrast, the ciphertexts from the proposed SF encryption scheme are presented as images and can be robust against distortion. We have verified that valid decryption can be achieved even if some parts of the images were missed, or some artifacts were introduced into the images. The advantage of good robustness is expected to increase the immunity to quality degradation in a practical information communication system.

In conclusion, we have proposed and experimentally demonstrated a dynamic optical encryption scheme that is promising for enhancing information security and capacity. We have shown that the SF of optical images can be modulated as information carriers with the developed RFO techniques. The experimental results indicated that various messages can be successfully encrypted into optical images and the ciphertext images show good security of the embedded information. Furthermore, by exploiting the position, profile, and time degrees of freedom of the SF, advanced coding rules can be established to increase the information capacity. In addition, we have shown that the dynamic encryption rate can be up to 500 Hz, which can be further improved. The proposed optical encryption scheme may hold promise for advancing the research of information encryption, storage, and processing.

Funding

Guangdong Basic and Applied Basic Research Foundation (No. 2023A1515030141, 2021A1515010257), National Natural Science Foundation of China (No. U23A20369), the Innovation Group Project of Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (No. SML2022007), the Leading Talents of Guangdong Province Program (No. 00201502).

CRediT authorship contribution statement

Haoyang Yu: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Min Ouyang:** Software, Methodology, Formal analysis. **Xin Zhang:** Resources, Funding acquisition. **Tianhua Feng:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Zhaohui Li:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.optcom.2024.130459.

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